

Use Case Specification: Book Event Ticket

System: Event Management System (EMS) | Component: Registration & Checkout

1. Preconditions and Postconditions

1.1 Preconditions

- The Attendee is authenticated with an active account and assigned the Attendee role (FR-01, SR-01).
- The event is published, active, not past its date, and has not been suspended by Platform Admins (FR-02, FR-03, FR-08).
- The event has available capacity remaining greater than or equal to the minimum purchase quantity (FR-02, FR-04, DR-04).
- The external payment gateway service and transactional email service are online and reachable via defined APIs (NFR-05, SR-03, SR-05).

1.2 Postconditions

- **Success Guarantees:**
 - The event's available capacity is reduced by the exact number of purchased tickets immediately (FR-04, NFR-03).
 - A unique digital ticket record with a globally unique UUID and scannable QR/barcode is created (FR-05, DR-01).
 - Payment transaction data is stored in tokenized form with masked card numbers, meeting PCI-DSS standards (NFR-02, DR-02).
 - An asynchronous confirmation email with the digital ticket receipt is dispatched via the transactional email service (SR-04, SR-05).
 - The Attendee is associated with the event and included in the Organizer's confirmed attendee list (FR-07, SR-06).
- **Minimal / Failure Guarantees:**
 - If payment authorization fails or the process is aborted, any temporary database locks are released immediately.
 - No tickets are generated, no inventory is deducted, and no customer charge is finalized (FR-04, NFR-03).

2. Main Success Flow (Basic Flow)

The primary flow describes the end-to-end journey of a user successfully finding an event,

reserving capacity, completing payment, and obtaining their digital ticket.

Step	Actor Action (Attendee)	System Response (EMS & Services)
1	The Attendee browses or filters active events by date and category on the public listings screen.	The system queries active events and presents the results in under 2 seconds (FR-03).
2	The Attendee selects an event to view full details (title, venue, schedule, ticket pricing, and remaining seats).	The system renders the event details page with a mobile-optimized viewport (NFR-04).
3	The Attendee selects the desired ticket quantity and clicks "Proceed to Checkout".	The system verifies that the requested quantity is $\leq$ remaining available capacity and acquires an ACID-compliant record lock at the database layer (SR-02, NFR-03).
4	The Attendee reviews the order summary and submits credit card details via the embedded payment gateway form.	The payment gateway directly intercepts payment fields, generates a secure transaction token, and transmits the authorization request over RESTful APIs (NFR-02, SR-03).
5	The Attendee awaits payment confirmation.	The external payment gateway approves the transaction and triggers a webhook callback to the EMS payment listener (SR-04).
6	—	The system updates ticket inventory immediately, decrementing available seats by the purchased count, and commits the database transaction (FR-04, NFR-03).
7	—	The system issues a digital ticket with a unique UUID and scannable QR code, storing the record and tokenized transaction reference (FR-05, DR-01, NFR-02).
8	The Attendee views the order confirmation screen.	The system displays the success screen and simultaneously invokes the transactional email service to send a receipt and PDF ticket asynchronously (SR-05, UR-05).

3. Alternate Flow: Payment Authorization Failure

This alternate path branches from **Step 5** of the Main Flow when the external payment gateway declines the transaction (e.g., due to insufficient funds, invalid security code, expired card, or

fraud trigger).

Step	Actor Action (Attendee)	System Response (EMS & Services)
5a	The Attendee awaits payment authorization.	The external payment gateway denies the charge and returns an authorization failure response code (e.g., "card_declined", "insufficient_funds") to the EMS webhook/API handler (SR-03, SR-04).
5b	—	The system executes a transaction rollback, immediately releasing the database lock on the reserved tickets and returning them to the general availability pool (SR-02, NFR-03).
5c	—	The system logs the unsuccessful attempt using tokenized references without persisting sensitive cardholder data, ensuring PCI-DSS compliance (NFR-02, DR-02). No ticket UUID is minted.
5d	The Attendee receives an on-screen failure alert.	The checkout screen presents a descriptive, user-friendly error message indicating that payment could not be processed and explaining the failure cause without revealing security-sensitive bank codes.
5e	The Attendee chooses to either re-enter payment details or cancel the checkout.	• If the Attendee updates card details and re-submits, the flow re-enters at Step 4 of the Main Flow. • If the Attendee abandons the transaction, the session is cleared and the user is redirected back to the event details page.